

RE020- Food Remedy Food Data Quality & Provenance

1. Introduction

Food Remedy relies on accurate food and product information to provide users with product details and personalized food recommendations. The existing Food Remedy recommendation architecture identifies information such as ingredients, allergens, dietary tags, nutrients, price and preparation time as important food-item data. The recommendation process also considers user information such as allergies, dietary requirements, health constraints and nutritional goals.

Because this information is used by the recommendation system, the quality of the underlying food data can directly affect the reliability of Food Remedy. Missing, incorrect, outdated or conflicting information may result in incorrect product information or inappropriate recommendations.

This issue has also been observed during Food Remedy's closed-beta testing. Testers reported cases where products were displayed without their associated information, the database did not show the correct allergies, some everyday products could not be found through the barcode scanner, and inaccurate database information affected recommendations.

The purpose of this research is to investigate the main food-data quality risks affecting Food Remedy and provide practical recommendations for improving the reliability, consistency and maintenance of the data used by the Backend and Database teams.

2. Food Data Requirements

The first step in improving data quality is understanding which information needs to be reliable. Based on the existing Food Remedy recommendation architecture, the main categories of food/product information include product identification, ingredients, allergens, nutritional information and dietary classifications.

Product identification

Product identification information allows Food Remedy to correctly identify and retrieve a product. Important information can include:

- Product ID
- Barcode

- Product name
- Brand
- Product category
- Package or serving size

A reliable identifier is particularly important for the barcode-scanning feature because the scanned barcode needs to correspond to the correct product record.

Ingredients and allergens

Ingredient information is important because it can be used to determine the contents of a product and identify potential allergens. Allergen information is particularly important because it can directly affect whether a product should be recommended to a particular user.

Food Standards Australia New Zealand (FSANZ) provides an important reference for allergen terminology and labelling in Australia. Current requirements specify allergens that must be declared when present, including wheat, fish, crustacean, mollusc, egg, milk, lupin, peanut, soy, sesame and specified tree nuts, among others.

Food Remedy should therefore maintain consistent allergen terminology and should distinguish between:

- Allergen confirmed as present
- Allergen confirmed as absent
- Allergen information unavailable
- Allergen information requiring verification

This distinction is important because missing information should not be treated as confirmation that an allergen is absent.

Nutritional information

Nutritional information may include:

- Energy
- Protein
- Carbohydrates
- Fat
- Saturated fat
- Sugar
- Fibre
- Sodium

- Serving size
- Measurement unit

The Australian Food Composition Database (AFCD) provides a useful example of structured food and nutritional data. It contains food details, nutrient profiles, sources and information about how nutrient values were derived

This demonstrates the importance of maintaining not only the nutritional value itself but also information about its source and derivation.

Dietary information

Food Remedy may also need dietary classifications such as vegan, vegetarian or gluten-free. These classifications should be represented consistently and should ideally be supported by ingredient information or a reliable source rather than being inferred only from the product name.

3. Data Quality Issues

There are several ways in which Food Remedy's food data could become unreliable.

The first is **missing information**. A product may exist in the database but has no ingredient list, allergen information, nutritional values or serving size. Missing information becomes particularly important when the missing field is required for a recommendation.

The second issue is **incorrect information**. For example, a product may have an allergen value recorded incorrectly or have an outdated nutritional value. Incorrect information is particularly difficult because the database may appear complete even though the information is wrong.

Inconsistent information is another problem. The same dietary category or allergen may be represented using different names or formats. For example, "Gluten Free", "gluten-free" and "GF" could be treated as different values if the database does not use standardized categories.

Duplicate records can also reduce data reliability. The same product may be stored more than once under slightly different names. If the duplicate records contain different nutritional or allergen information, the recommendation system may produce inconsistent results.

Another important issue is **outdated information**. Food products can change their ingredients, formulations, serving sizes and nutritional values. If Food Remedy continues using an older record, the information displayed to users may no longer represent the current product.

Finally, **conflicting information** can occur when different sources provide different values for the same product. For example, one source may indicate that a product contains milk while another source does not. Such conflicts should be identified rather than allowing the system to silently choose one value.

4. Evidence from Food Remedy Testing

The existing Food Remedy closed-beta feedback provides practical evidence that data quality is already affecting the application.

One tester reported that products were sometimes displayed without the information associated with them. Another tester reported problems with allergen detection and stated that the database was not showing proper allergies. There were also reports that common grocery products could not be scanned because they appeared not to exist in the database.

The feedback also connected these problems with recommendation quality. A tester reported that inaccurate databases and allergen information meant that recommendations were not always accurate. Another tester suggested that products containing a user's allergens should receive zero recommendation while suitable products should receive higher recommendation scores.

This shows that improving food-data quality is not only a database maintenance issue. It can affect several Food Remedy features at the same time:

Food data → Product information → Allergen detection → Recommendation accuracy

Improving the reliability of the underlying data should therefore be considered an important part of improving the overall reliability of Food Remedy.

5. Data Sources and Reliability

The reliability of food data depends not only on whether information is present but also on where the information came from.

Potential sources for Food Remedy may include:

- Manufacturer information
- Product labels
- Government or regulatory databases
- Established food databases
- Retailer information
- Third-party sources

- User-provided information

These sources should not necessarily be treated as equally reliable.

For example, current information obtained directly from a product manufacturer or product label may be more appropriate for branded product ingredients and allergens than unverified information from a third-party website.

Similarly, established nutritional databases can provide useful reference information. The AFCD, for example, provides nutrient information for foods commonly available in Australia and records information about the source and derivation of its data.

Food Remedy should therefore consider storing basic **data, provenance** information alongside important food records.

Data provenance information	Purpose
Data source	Identifies where information came from
Source reference	Allows information to be traced
Collection date	Shows when information was obtained
Verification date	Shows when information was last checked
Source type	Identifies manufacturers, databases, retailers, etc.
Data-quality status	Indicates whether information can currently be trusted

This would make it easier for the Backend and Database teams to identify questionable or outdated records.

6. Data Validation and Consistency

Food Remedy should use several types of validation rather than relying on a single check.

Basic validation should identify missing required fields and invalid formats. For example, a product should have a valid identifier and product name, while numerical nutritional values should follow appropriate formats and units.

Numerical values should also be checked for obviously invalid values. For example, negative values for calories, protein or serving size would indicate a data problem.

More importantly, Food Remedy should perform cross-field validation. This can identify problems that are not visible when individual fields are checked separately.

For example:

- **Ingredients:** Milk powder
Allergen: None

This record should be flagged because the two fields appear to conflict.

Another example would be:

- **Dietary classification:** Vegan
Ingredients: Milk

This should also trigger a data-quality review.

Duplicate detection should use product identifiers, barcodes, product names, brands and package sizes to identify records that may represent the same product.

Food Remedy should also track data freshness. A record that was collected or verified a long time ago may require review, particularly if it contains information that can change such as ingredients or allergens.

7. Data-Quality Requirements and Recommended Handling

The following table summarizes the main fields identified in this research and how they should be handled.

Data field	Quality requirement	Recommended validation	Handling if unreliable
Product ID	Unique and valid	Duplicate and format checks	Flag for review
Barcode	Correct and unique where applicable	Format and duplicate checks	Flag or correct
Product name	Complete and consistent	Required field and standardization checks	Standardized or review
Brand	Consistent	Normalization	Standardize
Ingredients	Accurate and current	Missing, source and cross-field checks	Review before affected recommendations
Allergens	Accurate and verified	Compare with ingredients and source	Prevent affected recommendation
Nutritional values	Valid numerical values	Range and format checks	Flag incorrect values
Serving size	Valid value and unit	Range and unit validation	Correct or flag
Dietary tags	Standardized and supported	Controlled values and ingredient checks	Review classification
Data source	Traceable	Source required for important data	Lower confidence/review

Collection date	Valid	Date validation	Flag missing information
Verification date	Sufficiently current	Freshness check	Mark for review
Data-quality status	Clearly defined	Apply validation results	Use to determine recommendation eligibility

The table shows that different types of data should receive different levels of protection. For example, a formatting issue with a product name is less serious than unreliable allergen information.

8. Handling Unreliable Data

Food Remedy should use a risk-based approach when handling unreliable data rather than treating every error in the same way.

A useful approach is to classify records into three levels:

Data status	Description	Recommended action
Verified	Complete, consistent and supported by a reliable source	Use normally
Needs Review / Uncertain	Minor missing information, older data or less certain source	Use cautiously or provide an uncertainty indication
Invalid / Unreliable	Critical information is missing, conflicting or known to be incorrect	Do not use it for affected recommendations

For example, a missing product description may not need to stop recommending. However, missing or conflicting allergen information should be treated much more seriously.

The recommendation system should therefore consider the reliability of the underlying food data before producing a result.

If critical data is unavailable, the system should **not assume a safe value**. For example:

- Missing allergen information ≠ No allergens

and:

- Missing nutritional value ≠ 0

This distinction is essential because treating missing information as a confirmed absence could create misleading recommendations.

9. Recommended Data-Quality Process

A practical process for Food Remedy could be structured as follows:

1. Product received: Food/product information enters the system.

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2. Product identification: Check product ID, barcode and possible duplicate records.

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3. Basic validation: Check required fields, formats, units and numerical values.

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4. Consistency checks: Compare related information such as ingredients, allergens and dietary classifications.

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5. Source and freshness check: Determine where the information came from and whether it is sufficiently current.

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6. Data-quality status: Assign a status such as:

- Verified
- Needs Review
- Uncertainty
- Invalid

↓

7. Recommendation decision: Use reliable information normally, provide appropriate uncertainty handling for less reliable information, and prevent recommendations where critical information cannot be trusted.

This approach also aligns with the existing Food Remedy research describing a **clean → enrich → seed** data pipeline.

10. Recommendations for Backend and Database Teams

Based on the findings, the following recommendations should be considered by the Backend and Database teams.

1. Prioritise critical food information

Ingredients and allergen information should receive the highest level of validation because errors in these fields can directly affect product suitability and recommendations.

2. Standardize important fields

Controlled values should be used for allergens, dietary categories, food categories and units. This will reduce inconsistent representations of the same information.

3. Track data sources and verification dates

Important records should include information about where the data came from and when it was last verified.

4. Introduce automated validation

Automated checks should identify missing fields, invalid numerical values, duplicate products and conflicts between related fields.

5. Add a data-quality status

Products should have a status indicating whether their information is verified, requires review or should not currently be used.

6. Introduce recommendation safeguards

The recommendation system should consider data quality before using a product. Critical unreliable information should prevent the affected recommendation rather than being treated as trustworthy.

7. Review outdated information

Records containing older ingredients, allergen or nutritional information should be identified for verification.

8. Use appropriate authoritative sources

Where external data is required, the team should evaluate recognized sources such as the Australian Food Composition Database for appropriate nutritional information and Australian food-labelling requirements as a reference for allergen and ingredient handling.

11. Priority of Recommendations

The recommendations can be prioritized based on their potential impact on Food Remedy.

Priority	Area	Reason	
High	Allergen and ingredient validation	Directly affects dietary suitability and recommendation safety	
High	Handling missing/conflicting critical data	Prevents unreliable recommendations	
High	Source and verification tracking	Improves trust and traceability	
Medium	Duplicate detection	Prevents conflicting product records	
Medium	Nutritional validation	Improves accuracy of nutrition-based recommendations	
Medium	Data freshness checks	Helps identify outdated product information	
Lower	Minor formatting standardization	Improves consistency but has less direct risk	

12. Conclusion

Food Remedy's food/product data is a critical component of its product information and recommendation features. Existing project documentation identifies ingredients, allergens, dietary tags and nutritional information as important inputs to the recommendation process.

The closed-beta feedback demonstrates that data-quality problems are already affecting the application, including incorrect allergen information, missing product information, limited product availability through barcode scanning and inaccurate recommendations.

The research identified **missing, incorrect, inconsistent, duplicate, outdated and conflicting data** as the major risks that should be addressed.

The recommended approach is not to treat every data issue equally. Minor formatting problems can generally be corrected or standardized, while critical issues involving ingredients, allergens or product identity should receive stronger validation and may need to prevent a recommendation.

Food Remedy would benefit from introducing a structured data-quality process that combines validation, standardization, duplicate detection, source tracking, freshness checks and data-quality statuses. This would provide the Backend and Database teams with a clearer method for identifying unreliable records and would reduce the likelihood of the recommendation system relying on inaccurate or incomplete information.

Ultimately, improving food-data quality should improve the reliability of **product information, allergen detection and personalized recommendations**, directly supporting the goal of making Food Remedy's recommendations more consistent and trustworthy.

13. Reference

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